PADHAKU
Curiosity · Class 6 Science



CHAPTER 4

Exploring Magnets

NCERT Exercise Solutions

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✨ Try each question yourself first, then check the answer. 💡 answers are samples: write your own.

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⭐ 4.1 Magnetic and Non-magnetic Materials

? Do magnets stick to objects made of certain materials only?

✔ ANSWER

Yes. Magnets stick only to objects made of certain materials, such as **iron**, nickel and cobalt.

They do not stick to objects made of wood, plastic, rubber, glass, paper or cloth.

Activity 4.1 **Bring a magnet near objects made of different materials one by one. Predict and record in Table 4.1 which objects stick to the magnet.**

Was your prediction correct for all objects? Which materials stuck to the magnet? What conclusion can you draw?

💡 SAMPLE ANSWER

Object	Material	Attracted? (Observation)
Pencil	Wood	No
Eraser	Rubber	No
Iron nail	Iron	Yes
Steel spoon	Steel (mostly iron)	Yes
Plastic scale	Plastic	No

Glass bangle	Glass	No

Only objects made of **iron** (and steel) stuck to the magnet.

Conclusion: Some materials are attracted by a magnet (**magnetic materials**) and others are not (**non-magnetic materials**).

? Do all parts of a magnet attract magnetic materials equally?

✔ ANSWER

No. A magnet attracts most strongly near its **ends (poles)**. The middle part attracts very little (see Activity 4.2).

? Which materials listed in Table 4.1 were found to be non-magnetic?

💡 SAMPLE ANSWER

Wood (pencil), rubber (eraser), plastic, glass, paper and cloth were **non-magnetic**. They were not attracted by the magnet.

⭐ 4.2 Poles of Magnet

Activity 4.2 **Spread iron filings on a paper and place a bar magnet over them. Tap the paper.**

Do the iron filings stick all over the magnet uniformly? Or do they stick more at some places?

✔ ANSWER

Observation: Most iron filings stick near the **two ends** of the bar magnet; very few stick in the middle.

Conclusion: The attraction is strongest at the ends of a magnet. These ends are called its **poles** (North pole and South pole).

? **If we repeat this activity with magnets of other shapes, do we get the same result?**

✔ ANSWER

Yes. With a magnet of any shape (U-shaped, ring etc.), most iron filings stick to its **poles**.

? **Can we find a magnet with a single pole?**

✔ ANSWER

No. It is **not possible** to get a magnet with a single pole.

Even if a magnet is broken into very small pieces, each piece has both a North pole and a South pole.

⭐ 4.3 Finding Directions

Activity 4.3 **Suspend a bar magnet by a thread tied at its middle. Rotate it gently and let it come to rest. Mark the line along which it rests. Rotate it again.**

Does the magnet rest along the same line?

✔ ANSWER

Yes. Every time, the magnet comes to rest along the **same line**.

? **What is this direction along which the magnet rests? How can we find it out?**

✔ ANSWER

The magnet always rests along the **north-south direction**.

We can find it from the direction where the Sun rises (East) or sets (West). If East is on your right, North is in front of you. The magnet's line points North–South.

Activity 4.3 **Repeat the activity with a small iron bar in place of the bar magnet. What do you observe? Does it always rest along north-south direction?**

✔ ANSWER

No. The iron bar can rest in **any direction**.

Only a magnet always rests along the north-south direction. So this is a way to test whether a piece of metal is a magnet or not.

? **How can we make our own magnetic compass?**

✔ ANSWER

Make an iron needle into a magnet by rubbing it with a magnet in one direction many times. Push it through a cork and float the cork on water. The needle comes to rest in the north-south direction (see Activity 4.4).

Activity 4.4 **Make an iron sewing needle a magnet by stroking it 30–40 times with one pole of a magnet. Pass it through a cork and float it on water in a bowl. Rotate the cork gently and wait till it stops. Repeat. Do the ends of the needle always point in the same direction?**

✔ ANSWER

Yes. The magnetised needle attracts iron filings/pins, so it has become a magnet.

Each time it comes to rest, the needle points in the **same direction (north-south)**, and the same end always points north.

Conclusion: Our home-made magnetic compass is ready to find directions.

⭐ 4.4 Attraction and Repulsion between Magnets

? **What happens when we bring two magnets closer to each other?**

✔ ANSWER

They either **attract** or **repel** each other.

Unlike poles (N–S) attract; like poles (N–N or S–S) repel.

Activity 4.5 **Place bar magnet A on 5–6 round pencils. Bring one end of magnet B near one end of A (without touching). Then bring the other end of B near the same end of A.**

Does magnet A begin to move? Does it always move in the direction of the approaching magnet? What do these observations suggest?

✔ ANSWER

Yes, magnet A moves on the pencils.

No, it does not always move towards magnet B. With one end of B it moves **towards** B (attraction); with the other end it moves **away** (repulsion).

Conclusion: Unlike poles attract each other; like poles repel each other.

Activity 4.5 **Repeat the activity by using an iron bar in place of one of the magnets. What do you observe this time?**

✔ ANSWER

Both ends of the iron bar are **attracted** by both the North and South poles of the magnet.

There is no repulsion.

So a magnet can be identified by its property of **repulsion**.

Activity 4.6 **The compass needle is also a magnet. Will it show the same behaviour if a magnet is brought closer to it?**

Bring the North pole of a bar magnet close to the North pole of the compass needle. Does the needle deflect? If yes, in which direction? Repeat with the South pole. Do you observe any difference?

✔ ANSWER

Yes, the compass needle behaves like a magnet.

With the **North pole** of the bar magnet, the North pole of the needle moves **away** (repulsion).

With the **South pole** of the bar magnet, the North pole of the needle moves **closer** (attraction).

? **Suppose we place a piece of wood between the compass needle and the magnet. Will this affect the deflection of the compass needle?**

✔ ANSWER

No. The needle still deflects in the same way. The magnet acts through wood.

Activity 4.7 **Place a piece of wood, then cardboard, thin plastic and thin glass sheet between the bar magnet and the compass. Is there any effect on the deflection? Record in Table 4.2.**

✔ ANSWER

S. no.	Material placed	Observation
1	Wood	Needle still deflects; no appreciable change
2	Cardboard	No appreciable change
3	Plastic	No appreciable change
4	Glass	No appreciable change

Conclusion: The magnetic effect can act **through non-magnetic materials**.

⭐ 4.5 Fun with Magnets

? **Can we make a garland? (Fig. 4.11)**

💡 SAMPLE ANSWER

Yes. Hang steel paper clips or pins one below the other from a magnet. Each clip holds the next one, making a magnetic garland.

? **Can we take the steel balls out of the maze by moving a magnet below the cardboard tray? (Fig. 4.12)**

✔ ANSWER

Yes. The magnet attracts the steel balls through the cardboard (a non-magnetic material), so moving the magnet below the tray moves the balls along the path.

? **Can we pick out a steel paper clip fallen in water using a magnet, without getting our fingers or the magnet wet? (Fig. 4.13)**

✔ ANSWER

Yes. Move a magnet along the outside of the glass/bowl. The clip follows the magnet up the side and can be taken out at the top. The magnetic effect acts through glass and water.

? **Will the two cars speed towards each other or run away from each other when brought**

closer? (Fig. 4.14 – like poles facing each other)

✔ ANSWER

They will **run away** from each other, because **like poles repel**.

Let us enhance our learning

1 (i) Unlike poles of two magnets _____ each other, whereas like poles _____ each other.

✔ ANSWER

attract; repel

1 (ii) The materials that are attracted towards a magnet are called _____.

✔ ANSWER

magnetic materials

1 (iii) The needle of a magnetic compass rests along the _____ direction.

✔ ANSWER

north-south

1 (iv) A magnet always has _____ poles.

✔ ANSWER

two

2 (i) **A magnet can be broken into pieces to obtain a single pole.**

✔ ANSWER

False (F). Every piece, however small, has both N and S poles.

2 (ii) **Similar poles of a magnet repel each other.**

✔ ANSWER

True (T). Like poles repel.

2 (iii) **Iron filings mostly stick in the middle of a bar magnet when it is brought near them.**

✔ ANSWER

False (F). They mostly stick near the ends (poles).

2 (iv) **A freely suspended bar magnet always aligns with the north-south direction.**

✔ ANSWER

True (T).

3 **Fill in the blanks (Column I: poles placed near each other; Column II: interaction).**

Column I	Column II
N–N	_____
N–___	Attraction
S–N	_____
___–S	Repulsion

✔ ANSWER

Column I	Column II
N–N	**Repulsion**
N–**S**	Attraction
S–N	**Attraction**
S–S	Repulsion

4 **Atharv rolled a bar magnet over a heap of steel U-clips. Which option is likely to be his observation? (A and C are the ends, B is the middle.)**

Option	A	B	C
(i)	10	2	10
(ii)	10	10	2
(iii)	2	10	10
(iv)	10	10	10

✔ ANSWER

Option (i): 10, 2, 10.

The attraction is strongest at the poles (ends A and C) and weakest in the middle (B).

5 **Reshma bought three identical metal bars. Two were magnets and one was a piece of iron. How will she identify the two magnets (without using any other material)?**

✔ ANSWER

Bring the ends of two bars near each other, then turn one bar around and try again.

- If at some position the two bars **repel**, both of them are magnets.
- If they only attract at both ends, one of them is the iron bar.

Repulsion happens only between two magnets, so the pair that repels are the magnets.

6 **You are given a magnet which does not have the poles marked. How can you find its poles with the help of another magnet which has its poles marked?**

✔ ANSWER

Bring the **North pole** of the marked magnet near one end of the unmarked magnet.

- If that end is **repelled**, it is the **North pole**.
- If it is **attracted**, it is the **South pole**.

The other end is the opposite pole.

7 **A bar magnet has no markings to indicate its poles. How would you find out near which**

end its North pole is located without using another magnet?

✔ ANSWER

Tie a thread at the middle of the magnet and hang it freely. Let it come to rest.

It rests along the north-south direction. The end pointing towards **north** (when you face the rising Sun, north is on your left) is its **North pole**.

8 **If the earth is itself a magnet, can you guess the poles of earth's magnet by looking at the direction of the magnetic compass?**

✔ ANSWER

Yes. The North pole of the compass needle points towards the geographic north.

Since unlike poles attract, the earth's magnet must have its **South pole near the geographic north** and its **North pole near the geographic south**.

9 **While a mechanic was repairing a gadget using a screwdriver, the steel screws kept falling down. Suggest a way to solve the problem.**

✔ ANSWER

Make the screwdriver a magnet: rub one pole of a magnet along the screwdriver from one end to the other, in the same direction, many times (as done with the needle in Activity 4.4).

The magnetised tip will **attract and hold** the steel screws, so they will not fall.

10 **Two ring magnets X and Y are arranged as shown (X above Y on a stand). Magnet X does not move down further. What could be the reason? Suggest a way to bring X in contact with Y, without pushing either magnet.**

✔ ANSWER

Reason: The facing sides of X and Y are **like poles**, so they repel each other and X stays up in the air.

Way: Take out X, turn it upside down and put it back. Now **unlike poles** face each other, so X is attracted and comes down to touch Y.

11 **Three bar magnets are arranged on a table as shown in Fig. 4.17. What is the polarity, N or S, at the ends 1, 2, 3, 4 and 6? End 5 is N.**

💡 SAMPLE ANSWER

Each magnet has N at one end and S at the other; ends that touch each other must be unlike poles (they attract).

- End 5 = N, so the other end of that magnet, **4 = S**.
- End 1 touches 4, so **1 = N**; the other end of that magnet, **2 = S**.
- End 3 touches 2, so **3 = N**; the other end, **6 = S** (and 6 touches 5 = N, which fits).

Answer: 1 – N, 2 – S, 3 – N, 4 – S, 6 – S (check the picture in your book)

📘 Learning further

1 **Using 3–4 different magnets, try to lift steel pins or U-clips and check which magnet picks up the largest number. Discuss why different magnets might have picked up different numbers of pins.**

💡 SAMPLE ANSWER

The magnets pick up different numbers of pins because they have different **strengths**. A stronger magnet (often bigger or made of a different material) picks up more pins.

2 **Make a toy 'Hopping Frog': fix ring magnets in alternate North-South fashion along a scale; glue a ring magnet under a paper frog and attach a plastic strip; slide it over the scale.**

💡 SAMPLE ANSWER

When the frog's magnet moves over a magnet with the **same pole** facing up, it is **repelled** and jumps up; over an **unlike pole** it is pulled down. So the frog keeps hopping as it slides.

3 **Find out about the Maglev Train and try to make its model.**

💡 **SAMPLE ANSWER**

Maglev means magnetic levitation. Strong magnets on the track and the train **repel** each other, so the train floats above the track and does not touch it. With less friction it can run very fast.

Model: stick magnets along a cardboard track and under a light paper train with like poles facing; the train floats and slides along.

4 **Try to find out why there is a need to make magnets of different shapes.**

💡 **SAMPLE ANSWER**

Different jobs need different shapes: a needle for a compass, a U-shape to make a strong pull between its two close poles, rings for speakers and toys, discs for fridge stickers and pencil boxes, bars for laboratory work.

5 **Collect information related to the use of magnets in the field of medicine.**

💡 SAMPLE ANSWER

- **MRI machines** use very strong magnets to take pictures of the inside of the body.
- Magnets are used to remove small iron pieces from the eye or wounds.
- Some hearing aids and medical devices use small magnets.

🎉 **Chapter 4 done! Revise the answers once more tomorrow.**